

ORGNZ-08: Grail Segments

Series: ORGNZ | **Notebook:** 8 of 10 | **Created:** January 2026 | **Last Updated:** 04/03/2026

Overview

Grail segments are logical groupings of data that enable real-time filtering across massive datasets without creating thousands of individual rules. Segments bring business context to observability data and are consistently available across the Dynatrace platform.

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS environment with Grail enabled
Permissions	<code>storage:filter-segments:read</code> , <code>storage:logs:read</code> , <code>storage:entities:read</code>
Knowledge	Completed ORGNZ-07 (Advanced Permissions)
Data	At least 1 hour of log, span, and entity data

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| **Dynatrace Environment** | SaaS environment with Grail enabled |

| **Permissions** | `segment:read` for viewing, `segment:write` for creating |

| **Knowledge** | Completed ORGNZ-01 through ORGNZ-07 |

Learning Objectives

- By the end of this notebook, you will:
- Understand segments vs buckets
 - Design effective segment rules
 - Use dynamic variables in segments
 - Apply entity relationships for comprehensive filtering

Segments vs Buckets

Aspect	Buckets	Segments
Purpose	Physical data storage	Logical data filtering
Retention	Controls data retention	No impact on retention
Data types	One type per bucket	Multiple types per segment
Access control	Can be used with IAM	Data filtering only
Cost impact	Direct storage costs	No direct cost impact
Query scope	Storage-level partitioning	Query-time filtering

Key insight: Buckets store data; segments filter it. Use buckets for retention and cost allocation, segments for dynamic data views.

Performance benefit: Applying a segment also reduces the **scanned bytes** of your queries. Since segments inject filter conditions at query time, only matching data is scanned — meaning fewer bytes read, lower cost, and faster results. This is especially impactful on high-volume buckets where unfiltered queries might hit the 500 GB scan limit.

Segment Use Cases

1. Team-Based Views

Segment: "Platform Team"

Includes: Infrastructure hosts, Kubernetes clusters, platform services

Benefit: Team sees only relevant data, faster troubleshooting

2. Application Isolation

Segment: "Checkout Application"

Includes: Checkout service, related databases, frontend components

Benefit: Complete application view across logs, traces, metrics

3. Environment Filtering

Segment: "Production Only"

Includes: Entities with environment=production tag

Benefit: Focused production monitoring without dev/test noise

Segment Structure

```
Segment:
  name: "segment-identifier"
  displayName: "Human Readable Name"
  description: "What this segment filters"

  includes:
    - type: logs
      filter: "host.name starts-with 'prod-'"

    - type: dt.entity.host
      filter: "tags contains 'environment:production'"

    - type: spans
      filter: "service.name == 'checkout'"

  variables:
    - name: environment
      values: ["production", "staging"]
```

Segment Limits

Limit	Value	Notes
Maximum includes (rules) per segment	20	Plan rule complexity accordingly
Include blocks per data/entity type	1	Cannot have multiple rules for same type
Expressions per filter condition	10	Maximum conditions per filter
Values per variable	10,000	Maximum variable values

Wildcard Restrictions

Pattern	Supported?	Notes
starts-with	Yes	Must have at least one preceding character

Pattern	Supported?	Notes
contains	No	Not supported in segment filter conditions
ends-with	No	Not supported in segment filter conditions

Important: The `contains` restriction applies to **signal data** filters (logs, spans, events). For **entity type** includes, `tags contains 'value'` is valid because it checks tag membership, not substring matching.

Segment Design Patterns

Pattern 1: Team-Based Segment

```
name: team-platform
displayName: "Platform Team"
description: "All infrastructure and platform services"

includes:
  - type: dt.entity.host
    filter: "tags contains 'team:platform'"

  - type: logs
    filter: "dt.host_group.id starts-with 'platform-'"

  - type: spans
    filter: "service.name starts-with 'platform-'"

```

Pattern 2: Application-Based Segment

```
name: app-checkout
displayName: "Checkout Application"
description: "Checkout service and dependencies"

includes:
  - type: dt.entity.service
    filter: "tags contains 'app:checkout'"

  - type: logs
    filter: "k8s.namespace.name == 'checkout'"

  - type: spans
    filter: "service.name starts-with 'checkout-'"
```

Pattern 3: Environment-Based Segment

```
name: env-production
displayName: "Production Environment"
description: "Production workloads only"

variables:
  - name: env
    values: ["production", "prod"]

includes:
  - type: dt.entity.host
    filter: "tags contains 'environment:${env}'"

  - type: logs
    filter: "dt.host_group.id starts-with 'prod-'"
```

Leveraging OneAgent Enrichments

OneAgent automatically enriches signals with metadata for precise filtering:

Enrichment	Source	Example Use
<code>dt.host_group.id</code>	Host group configuration	Filter by host group
<code>k8s.namespace.name</code>	Kubernetes namespace	Filter by namespace

Enrichment	Source	Example Use
<code>k8s.cluster.name</code>	Kubernetes cluster	Filter by cluster
<code>cloud.provider</code>	Cloud detection	Filter by AWS/Azure/GCP
<code>cloud.region</code>	Cloud region	Filter by region
<code>tags</code>	Entity tags	Filter by custom tags

Host Group Filtering Example

```
includes:  
  - type: logs  
    filter: "dt.host_group.id == 'prod-web-tier'"  
  
  - type: dt.entity.host  
    filter: "hostGroup == 'prod-web-tier'"
```

Entity Relationship Strategies

Top-Down (Host-Centric)

Start from hosts and include related entities:

```
includes:  
  - type: dt.entity.host  
    filter: "tags contains 'team:platform'"  
  
  - type: dt.entity.process_group  
    relationship: "runsOn dt.entity.host"  
  
  - type: dt.entity.service  
    relationship: "runsOnProcessGroup dt.entity.process_group"
```

Bottom-Up (Service-Centric)

Start from services and include supporting infrastructure:

```
includes:
  - type: dt.entity.service
    filter: "tags contains 'app:checkout'"

  - type: dt.entity.process_group
    relationship: "hostsService dt.entity.service"

  - type: dt.entity.host
    relationship: "runsProcessGroup dt.entity.process_group"
```

Testing Segments

Lab Exercise: Complete Exercises 1-2 in **ORGNZ-08 LAB** for hands-on practice with these concepts.

Segments and Access Control

Important: Segments are for **data filtering only**, not access control.

Function	Segments	IAM Policies
Filter data in queries	Yes	No
Restrict user access	No	Yes
Enforce security boundaries	No	Yes
Dynamic data views	Yes	No

Segments are governed by existing access controls - users only see data they're authorized to access.

Segment Design Checklist

- ☐ Defined clear segment purpose
- ☐ Identified all data types to include

- [] Verified metadata consistency (tags, host groups)
- [] Planned rule structure (max 20 includes)
- [] Considered entity relationships
- [] Tested filter conditions with DQL
- [] Documented segment usage
- [] Planned variable usage for flexibility

Next Steps

Continue with the ORGNZ series:

- **ORGNZ-09**: Enterprise Patterns
- **ORGNZ-10**: Advanced Segment Definitions

References

- [Grail Segments](#)
 - [Entity Model](#)
 - [Segment Limits](#)
 - [Segments blog post](#)
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